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Online Voting and Turnout: Evidence from St. Gallen

Course Paper

Data Science for Public Policy:
From Econometrics to AI

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1 Introduction

Political participation is declining in most Western Democracies which potentially undermines the legitimacy of democratic decisions. Policymakers aim to address this issue by implementing policies that reduce voting costs, with the expectation of increasing voter turnout (Goodman & Stokes, 2020, p. 1155). This paper analyses the effect of one such policy: internet voting. Internet voting has a considerable history in Switzerland, with initial trials conducted in the early 2000s (Plüss, 2023). Germann and Serdült (2017) analysed the turnout effect of these early trials and found no significant impact on voter turnout in the cantons of Geneva and Zurich. These experiments were eventually abandoned due to security concerns (Plüss, 2023). In 2023, a second phase of internet voting experimentation began, including trials in the canton of St. Gallen. Given the unresolved security issues, it is crucial to determine whether internet voting actually generates positive turnout effects. In other words, is it worth the risk?

While past research is not very encouraging, one argument in favour of restarting trials is that the population has become more digitally savvy and accustomed to performing important tasks online, such as banking and payments. This increased digital proficiency might lead to larger effects than observed in the past. In this paper, we empirically test the current promises of internet voting. Specifically, we will analyse the causal effect of internet voting on turnout in federal and cantonal referendums by exploring the ongoing pilot project in St. Gallen. Since the number of people allowed in taking part in the trials is limited, only some municipalities made internet voting available to their electorate. The resulting quasi-random within-canton variation can be exploited with a difference-in-difference approach (two-way fixed effects). Our approach allows us to rule out potential time-fixed confounders that affect turnout rates between municipalities (e.g., differences in turnout between rural and urban areas) as well as temporal variations that impact all municipalities equally (e.g., highly significant referendums that mobilize a large number of voters). Additionally, we address recent econometric concerns with staggered treatment adoption (see Roth, Sant'Anna, Bilinski & Poe, 2023).

Our main finding is that the introduction of internet voting in municipalities in the canton of St. Gallen had no effect on turnout in referendums. The results are robust to specifications with controls and time trends, and are in line with previous empirical research on online voting in Switzerland. Despite internet voting being utilized by a notable minority of voters, it does not appear to mobilize abstainers. Instead, some regular voters switch from traditional ballot or postal voting to internet voting. Therefore, further reducing the cost of voting through internet voting does not significantly increase turnout. It is important to note, however, that postal voting in Switzerland is already highly convenient and widely used (see Germann & Serdült, 2017, p. 2). Contrary to the negligible impact of internet voting, Luechinger, Rosinger and Stutzer (2007, p. 191) found that the introduction of postal voting increased

voter turnout in Switzerland significantly. Therefore, the turnout effects of internet voting might be more distinct in countries where postal voting is not available.

The remainder of this paper is structured as follows: Chapter 2 briefly summarizes the existing literature on voter turnout and internet voting. In Chapter 3 the data is summarized, and the empirical methods along with the model specifications are discussed and formalized. Thereafter, Chapter 4 provides the main empirical results as well as tests on the robustness of the results. Finally, the main findings are presented, and certain limitations of this paper are emphasized.

2 Literature Review

Classical voting theory suggests that reducing the costs of voting by making it more accessible should increase turnout (Aldrich, 1993; Blais, 2006, p. 116). Supporting this theory, Karp and Banducci (2000, p. 223) found that postal voting significantly increased turnout in Oregon. Similarly, Luechinger, Rosinger and Stutzer (2007, p. 191) showed that the introduction of postal voting in Switzerland increased voter turnout in federal ballots by an average of 4.1 percentage points.

Conversely, the evidence regarding the impact of internet voting on voter turnout is inconclusive. Goodman and Stokes (2020, p. 1164) found that internet voting increased voter turnout significantly by 3.5 percentage points in Ontario, Canada. Notably, the impact was even greater—doubling in magnitude—when internet voting was introduced in areas where postal voting was not available. On the other hand, Ehin, Solvak, Willemson and Vinkel (2022) found no effects on turnout in Estonia. They emphasized, however, that access to voting in Estonia was already very good prior to the introduction of widespread internet voting. Goodman and Stokes thus hypothesise that there might be “a ceiling on how much convenience can increase turnout” (2020, p. 1164).

In the case of Switzerland, two studies support this thesis. Germann (2021) found that for Swiss expatriate voters, for whom postal voting is not very convenient, the temporary suspension of internet voting decreased turnout by 4.1 to 6.4 percentage points. In contrast, Germann and Serdült (2017, p. 2) found no turnout effects for Swiss domestic voters (cf. Chapter 1). Germann and Serdült’s research is very similar to the paper and hand. However, two main differences may lead to different results: firstly, we focus on the second wave of internet voting expansion in Switzerland, while they analysed the first wave. Secondly, from a methodological standpoint, we can incorporate recent advancements in the difference-in-difference approach.

3 Methods and Data

3.1 Methodological Framework

The goal of this paper is to estimate the causal effect of the introduction of internet voting on turnout in municipalities in the canton of St. Gallen. As with any question of causal inference, we do not know what the outcome (here turnout) would have been in a certain municipality if internet voting was not available (Angrist & Pischke, 2009). Thus, we need a plausible counterfactual (Germann & Serdült, 2017, p. 4). Since internet voting is spreading gradually across municipalities in St. Gallen, we have a set of municipalities where internet voting is not available. Therefore, we can use this control group as a counterfactual and we can exploit the quasi-random variation in treatment assignment (internet voting) with a difference-in-difference approach. The strategy compares the difference in turnout before and after the policy change (introduction of internet voting) between municipalities where internet voting was introduced and those that were unaffected by the policy change (no internet voting). As shown in Table 1, 18 of the 75 municipalities in the canton of St. Gallen introduced internet voting at some point in 2023 or 2024. The other 57 municipalities belong to the control group.

Roth et al. (2023, p. 2221) emphasise that the main identifying assumption of the difference-in-difference approach is the parallel trends assumption. This assumption presumes that, without a policy change, the treatment group would exhibit the same mean change in outcomes as the control group. In our context, we must assume that, in the absence of internet voting, the average change in turnout would have been the same for municipalities that introduced internet voting as for those that did not. While we cannot directly test the parallel trends assumption, we can offer suggestive evidence supporting it by examining the pre-trends before the policy change. Figure 1 plots the turnout trends for municipalities that introduced internet voting later and those that belong to the control group prior to the introduction of internet voting. The turnout trends seem to be very similar, which indicates that the parallel trends assumption may be credible. Another identification assumption is the "no-anticipation assumption," which asserts that the treatment has no causal impact before its implementation (Roth et al., 2023, p. 2222). This means that the expectation of the introduction of internet voting does not influence voter turnout prior to its actual availability in elections. We have no evidence suggesting that this assumption is violated.

As a baseline specification, we thus estimate the following two-way fixed effects (TWFE) model:

$$Y_{it} = \alpha_i + \gamma_t + \beta Internet_Voting_{it} + \epsilon_{it} \quad (1)$$

where Y_{it} is the voter turnout in a given municipality i in time t ; α_i is the municipality fixed effect; γ_t a time (referendum day) fixed effect that controls for common shocks and ϵ_{it} an idiosyncratic error term. The variable of interest is $Internet_Voting_{it}$: A dummy variable that is set to one if internet

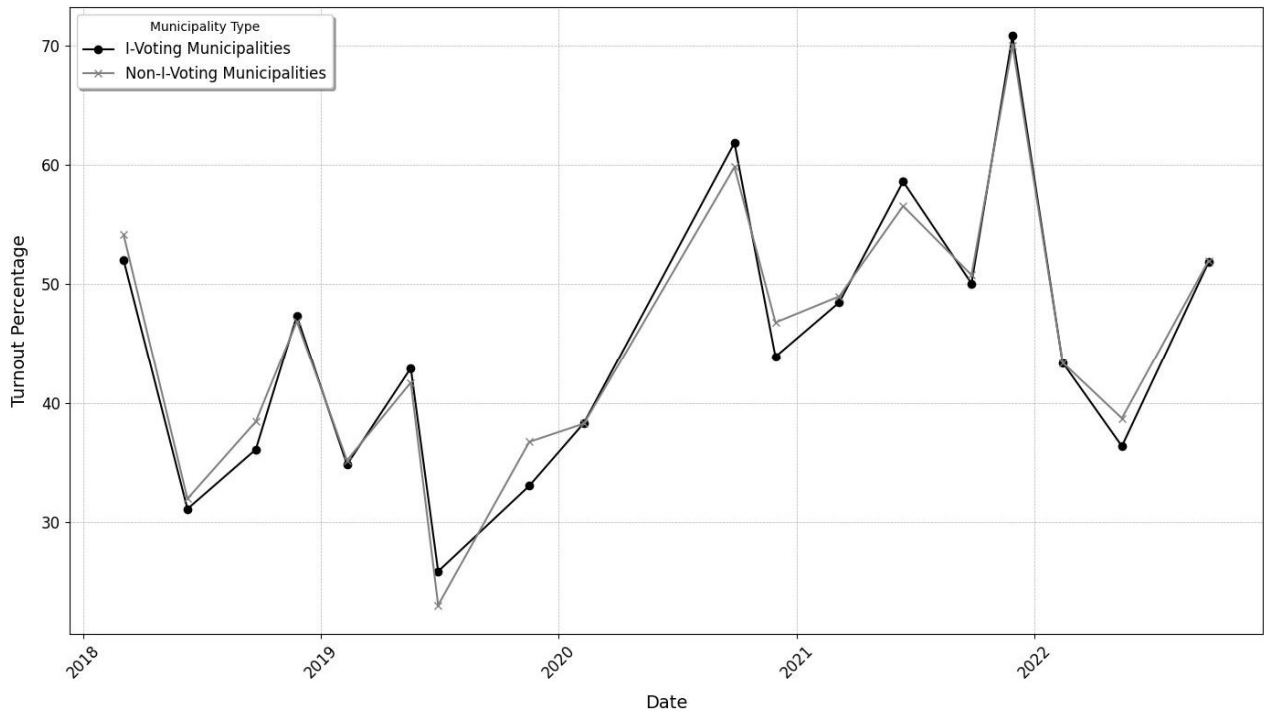
voting is available. Since we have municipality fixed effects, there is no need to control for municipality-level confounders that are fixed over time such as geographic, demographic or political differences that are constant over time. Furthermore, the time fixed effects ensure that we account for variations in turnout due to the significance of the referendums, provided these effects remain constant across the municipalities (Germann & Serdült, 2017, p. 5).

This specifications is, however, not robust to potential unobservable time-varying confounders (see e.g. Goodman & Stokes, 2020, p. 1161). To account for this possibility, we also estimate specifications including municipality-level time trends and some explicit control variables. Namely, demographic controls that might change over time (youth and old-age dependency ratios) as well as controls for the local political sentiment (party strength by municipality). The fully specified model then is:

$$Y_{it} = \alpha_i + \gamma_t + \varphi Z_{it} + \beta Internet_Voting_{it} + \delta X_{it} + \epsilon_{it} \quad (2)$$

where X_{it} is a matrix of covariate controls and Z_{it} the municipality-level linear time trends.

Figure 1
Pre-Trends.



3.2 Staggered Treatment and Heterogenous Effects

As mentioned earlier, not all municipalities introduced internet voting (were treated) at the same time. We therefore have a setting with staggered treatment. Braghieri, Levy & Makarin point out that even though TWFE regressions “are the workhorse models for staggered adoption research designs” they may not result in correct average treatment estimates if treatment effects are heterogenous (2022, p. 3672). According to Roth et al. (2023, p. 2219) this is because the difference-in-difference approach

combines two types of comparisons: clean comparisons between treated and not-yet treated units as well as forbidden comparisons between newly and already-treated units. If treatment effects are homogenous this is not problematic and we estimate consistent average treatment effects with the standard TWFE model (Braghieri et al., 2022, p. 3673). However, if the treatment effects are heterogeneous we might get biased estimates even if the parallel trends assumption holds (Roth et al., 2023, p. 2219). This could be the case, for instance, if the effects evolve gradually over time due to an initial period of cautious observation before widespread adoption of the new voting tool. Therefore, we reestimate our specifications using a robust estimator similar to the one introduced by Sun and Abraham (2021). The basic idea of this estimator is to only pick clean comparisons and get rid of the forbidden ones (Braghieri et al., 2022, p. 3673).

The specifications up to this point have focused on estimating an average treatment effect. However, it is quite plausible that the effects are heterogeneous. Germann and Serdült (2017, p. 8) hypothesize that internet voting might affect turnout differently across age groups. One of the primary promises of introducing internet voting is its potential to increase turnout among younger people, who typically abstain more frequently. While we would like to test this hypothesis, we unfortunately do not have access to turnout data segregated by age groups. Secondly, as Luechinger, Rosinger and Stutzer note in the context of the introduction of postal voting, reducing voting costs is expected to increase turnout more in municipalities that start from a low participation level as “there is simply a larger pool of non-voters that is potentially lifted over the participation threshold by a reduction in transaction costs” (2007, p. 188). Therefore, we will estimate specifications that test whether the effects are larger for low turnout municipalities.

3.3 Data

Table 1 presents the summary statistics for our dataset. We obtained the turnout data for each municipality and each referendum from the official website of the Canton of St. Gallen. The average turnout for the 5,175 observations (comprising 69 referendums across 75 municipalities) prior to the introduction of internet voting was 46.6%. Notably, turnout was slightly higher in municipalities that later adopted internet voting. Since the parallel trends assumption necessitates only that the control and treatment groups experience the same mean change in outcomes, rather than the same initial level, this difference does not compromise our identification strategy. Moreover, this observation mitigates potential concerns that municipalities with historically low turn-out were more inclined to adopt internet voting earlier. In addition, we collected demographic data (youth and old-age dependency ratios) and information on party strength by municipality from the Statistical Office of the Canton of St. Gallen. These data serve as controls for robustness checks. Table 1 provides summary statistics for

these variables for the year 2019. We can see that internet voting and non-internet voting municipalities display statistically significant differences in some of these characteristics. If these differences are not constant over time, it might be beneficial to include them as controls, as municipality fixed effects would no longer account for this variation.

Table 1
Summary Statistics.

	No I-Voting	I-Voting	Difference
Municipalities	57	18	
Referendums	69	17	
Average Turnout	0.463 (0.124)	0.477 (0.122)	-0.014*** (0.002)
Youth Dependency Ratio	0.357 (0.045)	0.356 (0.044)	0.001*** (0.000)
Old-Age Dependency Ratio	0.304 (0.056)	0.309 (0.051)	-0.005 (0.005)
SVP Voter Share	0.355 (0.061)	0.333 (0.048)	0.022 (0.014)
SP Voter Share	0.098 (0.039)	0.110 (0.045)	-0.013* (0.007)

Notes: All summary stats (except for average turnout) are from the year 2019, i.e. before the introduction of internet voting. Party voter shares are from the national elections. The column Difference reports *** 1% , ** 5% , * 10% significance.

4 Results

Table 2 summarizes our main results. Column 1 presents the effect of internet voting on voter turnout using a TWFE model with municipality and referendum day fixed effects. The coefficient indicates that the availability of internet voting is associated with a 0.43 percentage point increase in turnout holding constant all time-invariant characteristics across municipalities and cantonal trends. However, with a p-value of 0.326 we cannot reject the null hypothesis (no effect). Furthermore, the magnitude of the effect is also not meaningful. Next, we run a model that includes municipality-level linear time trends. The coefficient for internet voting remains quite robust to this specification and is still statistically insignificant. Column 3 reports the results after including the additional demographic and political covariates into the model. Again, the coefficient on internet voting remains consistent with our baseline specification. Although the focus of our analysis is on the effect of internet voting, it is worth noting that the coefficients for the demographic controls are statistically significant. Specifically, a higher proportion of older individuals in the population is associated with increased turnout, while a higher proportion of younger individuals is associated with decreased turnout (for further details, see the replication package).

Table 2

Main Results and Robustness Tests.

	(1)	(2)	(3)	(4)	(5)	(6)
Internet Voting	0.0043 (0.004)	0.0047 (0.004)	0.0049 (0.004)			
I-Voting (Early Cohort)				0.0047 (0.006)	0.0049 (0.006)	0.0049 (0.006)
I-Voting (Late Cohort)				0.0036 (0.003)	0.0044 (0.003)	0.0049 (0.003)
Municipality FE	✓	✓	✓	✓	✓	✓
Referendum Day FE	✓	✓	✓	✓	✓	✓
Time Trends		✓	✓		✓	✓
Controls			✓			✓
Robust Estimator				✓	✓	✓
Treated Municipalities	18	18	18	18	18	18
Untreated Municipalities	57	57	57	57	57	57
Referendums	86	86	86	86	86	86

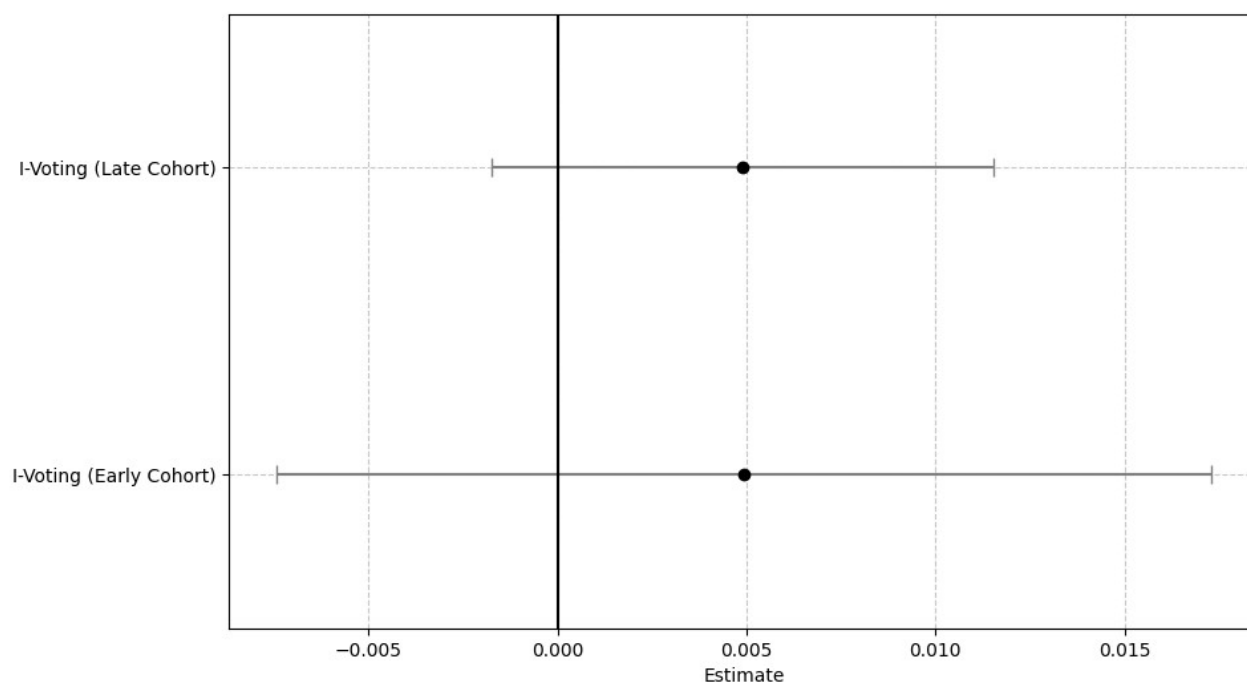
Note: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

As discussed in Chapter 3, the TWFE models displayed in Columns 1-3 assume homogeneity in treatment effects. To relax this assumption, we reestimate these specifications with a modified TWFE approach that allows for treatment effect heterogeneity. The results are displayed in Columns 4-6 of Table 2. We show separate treatment effects for the municipalities that adopted internet voting earlier versus those that adopted it later. However, the magnitude of the estimated treatment effects are fairly similar across the two cohorts. Moreover, they are also very similar to the results obtained by the classic TWFE estimations. For the fully specified model, we estimate a 0.49 percentage point increase in turnout with internet voting for both cohorts. As illustrated in Figure 2, these results are once again statistically indistinguishable from zero. Overall, these robust findings suggest that treatment effect heterogeneity is not a major concern in this context.

Lastly, we test the hypothesis that the effects of internet voting might be higher in municipalities that have, on average, lower participation levels. The results of the fully specified model are summarized in Table 3. The positive interaction term suggests that the introduction of internet voting has a higher effect—by approximately one percentage point—in low-turnout municipalities. In contrast, the coefficient for high-turnout municipalities is negative, suggesting that the overall positive effect is driven primarily by low-turnout municipalities. It is important to note, however, that these results are not statistically significant, with the exception of the interaction term in the late cohort.

Figure 2

Coefficient Plot of Fully Specified Model.

**Table 3**

Heterogeneity of Effects by Turnout Level.

Heterogeneity in Treatment Effects	
I-Voting (Early Cohort)	-0.0035 (0.006)
I-Voting*Low Turnout (Early Cohort)	0.0106 (0.007)
I-Voting (Late Cohort)	-0.0015 (0.004)
I-Voting*Low Turnout (Late Cohort)	0.0085*** (0.003)
Municipality FE	✓
Referendum Day FE	✓
Time Trends	✓
Controls	✓
Robust Estimator	✓

Note: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

5 Conclusion

Overall, our findings indicate that the introduction of internet voting has not led to a significant increase in turnout in the canton of St. Gallen. These results are robust across different model specifications, reinforcing the conclusions drawn in previous studies about internet voting in Switzerland. While this study contributes to the ongoing discussion about internet voting, it is important to consider its limitations. First, the effects of internet voting on different subgroups, particularly younger voters, were not investigated. Second, the second wave of internet voting adoption in Switzerland is still in its early stages. Consequently, the results presented here primarily reflect the short-term effects on turnout. It is possible that, in the long run, internet voting may become more widely adopted and mobilize a larger number of voters. Exploring these potential long-term effects presents an interesting avenue for future research.

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